

SEDAR 41 South Atlantic Red Snapper and Gray Triggerfish

Red Snapper

Data Workshop

Life History

Red Snapper Mini Season

If this program, along with continued closure of the fishery, is to extend into future seasons, an exploration of methods to further incentivize angler participation would be useful. After brief interviews with participants from the recreational fishers group at SEDAR 41, the following suggestions were provided to increase angler participation:

- Free fish cleaning at donation site.
- As people may be tired after being out on the water all day and with busy boat ramps, short questionnaire from a biologist on-site could be used instead of the anglers filling the forms out or requiring fishermen to fill out a survey online after they return home.
- Advertise data collection at local bait & tackle shops.
- Use NOAA's announcement system on weather radio channel where they also announce season closures, etc. Since fishermen are frequently monitoring this channel for weather updates, it could be an effective communication route to announce the collection information (drop locations, reward information, etc.).
- Dry storage areas are a good place to sample; many people store boats there instead of trailering home.

Life History Research

- More research on red snapper movements and migrations in Atlantic waters is needed. Available data and the results of studies in the Gulf of Mexico indicate high site fidelity, but that tropical storms may cause greater than normal movement that might help dispersal to depleted areas. This needs to be confirmed in the South Atlantic. Additional acoustic and traditional tagging is needed on known spawning locations to document spawning migrations or aggregations, and return of fish to non-spawning areas.
- Evaluate more thoroughly the data/sample collection during the mini-season to improve utility for assessments. This should include what samples should be collected (e.g. reproductive information).
- Possible changes in life history parameters, in particular relative to reproduction, need to be further investigated.
- Much is unknown about the early life history of Red Snapper, in particular relative to spawning areas, larval and juvenile stages, including habitat and dispersal.
- Alternative methods of reproductive output. The methods described in Klibansky's SEDAR41-DW49 may provide a more accurate estimate of reproductive output than previously used.

Further investigation into this modeling effort and use for future assessments should be investigated.

- Duration of spawning indicators. The definition of spawning indicators has received significant discussion recently. As this has significant implications for the estimates of reproductive output, further research is needed to define consistent criteria for spawning indicators in finfish.
- Continuing the age reading comparisons and calibrations between labs on a reference collection of known age fish would be beneficial for determining a more accurate aging error matrix and would provide accuracy to the age composition data.

Commercial Statistics

Landings

- Improve gear and effort data for each trip.
- Standardize methodology for developing average proportions to parse out unclassified landings.

Discards

- Investigate the validity and magnitude of “no discard” trips. This may include fisher interviews throughout the region.
- Examine potential impacts of “no discard” trips on estimated discards.
- Improve discard logbook data collections via program expansion or more detailed reporting (i.e. electronic logbooks, etc.)
- Establish an observer program that is representative of the fisheries in the South Atlantic

Biosampling

- Establish an observer program that is representative of the fisheries in the South Atlantic.
- Angler education with regards to recording depths on paper logbooks (i.e. standardized units); validation of additions to the logbook form still needed.
- Standardize TIP sampling protocol to get representative samples at the species level.
- Standardize TIP data extraction.

Recreational Statistics

- Complete analysis of available historic photos for trends in CPUE and mean size of landed Red Snapper and Gray Triggerfish for pre-1981 time period. (Ultimately all species).
- Formally archive data and photos for all other SEDAR target species.
- For Hire Survey (FHS) should collect additional variables (e.g. depth fished).
- Increasing sample sizes for at-sea headboat observers (i.e. number of trips sampled).
- Compute variance estimate for headboat landings.
- Mandatory logbooks for all federally permitted for-hire vessels.

Indices

- Compare existing methods and/or develop new methods to define effective effort in fishery dependent data.

- Estimate selectivity of video gear in the SERFS.
 - Tagging, stereo cameras
- For video reading, evaluate methods to score water clarity and habitat.
- Evaluate effect of (non) independence between chevron traps and videos, including methods to combine the indices.
- Continue exploring the use of continuous predictor variables (e.g., splines or polynomials) for ZIP and ZINB standardization models.
- Headboat at-sea observer program needs depth data from all states (not just FL) and increased coverage overall.
- SCDNR charterboat logbook program should be replicated by other states.
- Develop fishery independent hook-gear index (S41-DW08).

Assessment Workshop

- Increased fishery independent information, particularly maintaining reliable indices of abundance and composition data streams.
- Red Snapper were modeled in this assessment as a unit stock off the southeastern U.S. For any stock, variation in exploitation and life-history characteristics might be expected at finer geographic scales. Modeling such sub-stock structure would require more data, such as information on the movements and migrations of adults and juveniles, as well as spatial patterns of larval dispersal and recruitment. In addition, it is unclear whether a spatial model would improve the assessment.
- More research to describe the juvenile life history of Red Snapper is needed, including more work to identify the location of juveniles before they recruit to the fishery.
- The effects of environmental variation on the changes in recruitment or survivorship.
- The Florida sampling program, during the miniseason in particular, provided invaluable data to this assessment. Programs such as these would be useful in all South Atlantic states, particularly if the management regulations continue to make established methods of index development or composition sampling from fleets less regular or possible.

Review Workshop

The Review Panel considers the first three of the following bullets to be the highest priority for assessment improvement.

- Increased fishery independent information, particularly maintaining reliable indices of abundance and composition data streams.
- Improve the reliability of discard data as an abundance index by improving knowledge of private recreational fisherman behavior.
- Research to determine the spatial distribution (horizontal and vertical) of large adult Red Snapper using tracking and telemetry.
- The Review Panel reiterates various research recommendations focused on Red Snapper population structure in the South Atlantic. Red Snapper were modeled in this assessment as a

unit stock off the southeastern U.S. For any stock, variation in exploitation and life-history characteristics might be expected at finer geographic scales. Modeling such sub-stock structure would require more data, such as information on the movements and migrations of adults and juveniles, as well as spatial patterns of larval dispersal and recruitment, and spatially-explicit data of all types used in the assessment model. It is unclear whether a spatially-explicit model would improve the assessment. Given the robust ocean circulation in the South Atlantic Bight conditions creating population sub-structure. The research effort necessary to support such an effort would be extensive and probably unjustified on stock assessment improvement grounds, however, it would be needed to support MPA placement, performance evaluation, etc.

- More research to describe the juvenile life history of Red Snapper is needed, including more work to identify the location of juveniles before they recruit to the fishery.
- The effects of environmental variation on the changes in recruitment or survivorship.
- Investigate possible historical changes in sexual maturity. The current estimate of age of sexual maturity is low and unusual for other Lutjanids. Is it right or a compensatory response to heavy exploitation?
- Continue conducting studies to develop a time series of batch fecundity to obtain information on the inter-annual variation in reproductive output.

Gray Triggerfish

Data Workshop

Life History

- More research on gray triggerfish movements and migrations in Atlantic waters is needed. Available data and the results of studies in the Gulf of Mexico indicate high site fidelity, but that tropical storms may cause greater than normal movement that might help dispersal to depleted areas. This needs to be confirmed in the South Atlantic. Additional acoustic and traditional tagging is needed on known spawning locations to document spawning migrations or aggregations, and return of fish to non-spawning areas.
- Age validation study that should include edge type and the potential for using various age structures for use in assessment. This should include the logistical feasibility of using these alternative structures for routine sampling and processing.
- Early life history is largely unknown. E.g. size and age at settlement and length of the pelagic stage.
- Estimates of delayed bycatch mortality are needed. This should include the effect of cloacal protrusion as a result of barotrauma.
- Tagging studies are needed to define spawning locations (only shelf edge or not) and, movement, the results of which could be used to help inform fishing mortality and natural mortality.
- Impact of climate change on mortality and recruitment.

- Research on spawning behavior/nesting and how it impacts survivorship and stock productivity.
- Determine fecundity type and estimate annual fecundity in Atlantic waters.
- Alternative methods of reproductive output. The methods described in Klibansky's SEDAR41-DW49 may provide a more accurate estimate of reproductive output than previously used. Further investigation into this modeling effort and use for future assessments should be investigated.
- Duration of spawning indicators. The definition of spawning indicators has received significant discussion recently. As this has significant implications for the estimates of reproductive output, further research is needed to define consistent criteria for spawning indicators in finfish.
- Investigate gray triggerfish competition for nests. The presence of competition for nest space may affect, among other things, the spawning success (reproductive output) and the choice of a spawner recruit relationship. Further investigation into the nesting behavior of gray triggerfish is needed to provide information to address these issues.

Commercial Statistics

Landings

- Require species level reporting in state trip ticket programs. Some states in process of instituting species level reporting for all species.
- Improve gear and effort data collections.

Discards

- Investigate the validity and magnitude of "no discard" trips. This may include fisher interviews throughout the region.
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- Improve discard logbook data collections via program expansion or more detailed reporting (e.g. more detailed logbook, electronic reporting)
- Establish an observer program that is representative of the fisheries in the South Atlantic.

Biosampling

- Standardize TIP sampling protocol to get representative samples at the species level.
- Standardize TIP data extraction.
- Establish an observer program that is representative of the fisheries in the South Atlantic.
- Increase untargeted sampling in NE and Mid-Atlantic observer programs.
- Increase untargeted dockside sampling in NE and Mid-Atlantic.

Recreational Statistics

- Complete analysis of available historic photos for trends in CPUE and mean size of landed Red Snapper and Gray Triggerfish for pre-1981 time period. (Ultimately all species).
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Assessment Workshop

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- Increased age sampling and evaluation of ageing error over the stock area and from all fleets, particularly the general recreational fleet.
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- More research to better understand the life history of Gray Triggerfish is needed, including natural mortality, maturity, and reproductive potential, particularly for the youngest ages.
- The effects of environmental variation on the changes in recruitment or survivorship.

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- The effects of environmental variation on the changes in recruitment or survivorship of Gray Triggerfish.